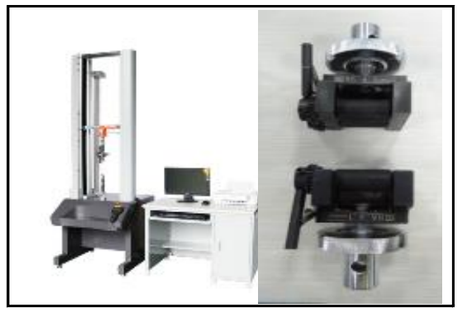
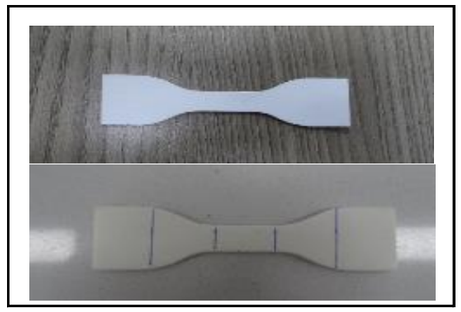
STANDARD OPERATING PROCEDURE OF TENSILE FOR RUBBER VF-B01

Lab /SOP/ 04-03



1. Tensile testing machine with chart recorder
2. Clamp for rubber ,EVA,PU
3. Thickness gauge
4. Steel scale or calipers



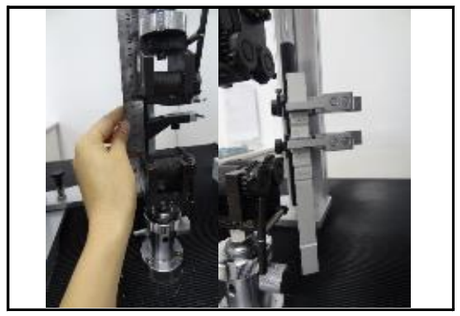
Cut 3 samples per material. the test samples must be of a uniform thickness particularly on the area of elongation bench mark - thickness range at 3.0+/- 1.0mm.
The cutting die should be dumb bell shaped, with a width of 1.0 +/-0.10cm in the gage area. draw two elongation bench marks at 2 cm apart



Measure and record the width of the test sample to the nearest 0.5mm (W)
Measure the thickness of the sample at each of the elongation bench marks and also halfway between the two bench marks.
Record the thinnest of the measurements to the nearest 0.1mm (T)



Zero and calibrate the tensile testing machine.



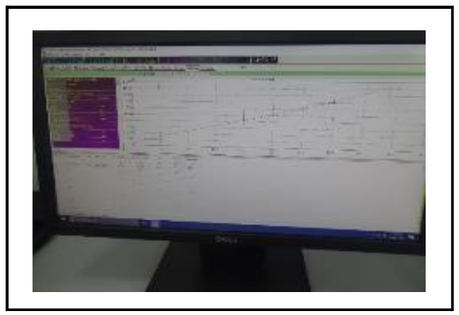
Set the tensile testing machine to the specified load range, crosshead speed and jaw gap.
Crosshead speed =500 mm/minute
Jaw gap =6.0+/- 0.3cm
Elongation bench mark gaps =2.0 +/- 0.1cm



Insert the sample centrally in the top and bottom jaws of the tensile testing machine
the sample is vertical.No slack or tension is present
Operate the tensile testing machine at the specified crosshead speed.



Record the breaking load to the nearest 0.1kgf (B)



Caculation:
Materials of unspecified thickness (rubber, plastic, foam etc.)
Tensile strength (kgf/cm2) = B / W x T
Elongation (%) = (D2- D1) / D1 x 100
Modulus (kgf/cm2) = M /W x T
Tensile strength is reported in kgf/cm2.

Prepared by	Approved by	REVISION		DATE	